

STAC Deep Dive

SpatioTemporal Asset Catalog specification and software ecosystem

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Element 84

What is it?

SpatioTemporal Asset Catalog

- Defined metadata for geospatial catalogs and assets
  - Focus on search and discovery
  - Simple and extensible
- Static catalogs
  - Crawlable
- Dynamic APIs
  - Searchable on arbitrary fields

The specification

- Catalogs
  - There's always a root catalog
  - Contains other catalogs and collections
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Ultimately contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when and where)

Item Fields

Field Name	Type	Description
id	string	**REQUIRED.** Provider identifier. As most geospatial assets are already defined by some identification scheme by the data provider it is recommended to simply use that ID. Data providers are advised to include sufficient information to make their IDs globally unique, including things like unique satellite IDs.
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
geometry	GeoJSON Geometry Object	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Specified in Longitude/Latitude based on EPSG:4326.
bbox	[number]	**REQUIRED.** Bounding Box of the asset represented by this item. Specified in Longitude/Latitude based on EPSG:4326 - first two numbers are longitude and latitude of lower left corner, followed by longitude and latitude of upper right corner. This field enables more naive clients to easily index and search geospatially. Most software can easily generate them for footprints. STAC compliant APIs are required to compute intersection operations with the item's geometry field, not its bbox.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is required.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key. Some pre-defined keys are listed in the chapter 'Asset types'.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type below). This field is *required* if such a relation type is present. This field provides an easy way for a user to search for any Items that belong in a specified Collection.

Item Properties

Field Name	Type	Description
datetime	string	**REQUIRED.** The searchable date and time of the assets, in UTC. It is formatted according to RFC 3339, section 5.6.
license	string	Item's license(s) as a SPDX License identifier or expression or `proprietary` if the license is not on the SPDX license list. Proprietary licensed data SHOULD add a link to the license text, see the `license` relation type.
providers	[Provider Object]	A list of providers, which may include all organizations capturing or processing the data or the hosting provider. Providers should be listed in chronological order with the most recent provider being the last element of the list.
title	string	A human readable title describing the item.
created	string	Creation date and time of this metadata file. This is NOT the timestamp the asset was created. MUST be formatted according to RFC 3339, section 5.6.
updated	string	Date and time this metadata file was updated last. This is NOT the timestamp the asset was updated last. MUST be formatted according to RFC 3339, section 5.6.

Item Assets

Field Name	Type	Description
href	string	**REQUIRED.** Link to the asset object. Relative and absolute links are both allowed.
title	string	The displayed title for clients and users.
type	string	Media type of the asset (see chapter on Media Types below).

Not Secure | landsat-stac.s3.amazonaws.com/catalog.json

Apps | Devseed | Beachfront | Gaming | Imported | sort | Geospatial | Remote Sensing | Screens | Programming

+ - View source


```
{
   id: "landsat-stac",
   stac_version: "0.6.0",
   description: "STAC for Landsat data",
 - links: [
    - {
         rel: "self",
         href: "https://landsat-stac.s3.amazonaws.com/catalog.json"
      },
    - {
         rel: "root",
         href: "./catalog.json"
      },
    - {
         rel: "child",
         href: "landsat-8-l1/catalog.json"
      }
   ]
}
```


inks

Content Extensions

Extension Name (Prefix)	Scope	Description	Maturity
Checksum (`checksum`)	Item +Catalog +Collection	Provides a way to specify file checksums for assets and links in Items, Catalogs and Collections.	*Proposal*
Data Cube (`cube`)	Item	Data Cube related metadata, especially to describe their dimensions.	*Proposal*
Datetime Range (`dtr`)	Item	An extension to provide datetime ranges with a start and an end datetime stamp in a consistent way.	*Proposal*
EO (`eo`)	Item	Covers electro-optical data that represents a snapshot of the earth for a single date and time. It could consist of multiple spectral bands, for example visible bands, infrared bands, red edge bands and panchromatic bands. The extension provides common fields like bands, cloud cover, off nadir, sun angle + elevation, gsd and more.	*Pilot*
Label (`label`)	Item	Items that relate labeled AOIs with source imagery	*Proposal*
Point Cloud (`pc`)	Item	Provides a way to describe point cloud datasets. The point clouds can come from either active or passive sensors, and data is frequently acquired using tools such as LiDAR or coincidence-matched imagery.	*Proposal*
SAR (`sar`)	Item	Covers synthetic-aperture radar data that represents a snapshot of the earth for a single date and time.	*Proposal*
Scientific (`sci`)	Item	Scientific metadata is considered to be data that indicate from which publication data originates and how the data itself should be cited or referenced.	*Proposal*

EO Extension

Field Name	Type	Description
eo:gsd	number	**REQUIRED.** Ground Sample Distance at the sensor.
eo:platform	string	**REQUIRED.** Unique name of the specific platform the instrument is attached to. For satellites this would be the name of the satellite (e.g., landsat-8, sentinel-2A), whereas for drones this would be a unique name for the drone.
eo:constellation	string	Name of the constellation that the platform belongs to. See below for details.
eo:instrument	string	**REQUIRED.** Name of instrument or sensor used (e.g., MODIS, ASTER, OLI, Canon F-1).
eo:bands	[Band Object]	**REQUIRED.** This is a list of the available bands where each item is a Band Object.
eo:epsg	integer|null	EPSG code of the datasource, `null` if no EPSG code.
eo:cloud_cover	number	Estimate of cloud cover as a percentage (0-100) of the entire scene. If not available the field should not be provided.
eo:off_nadir	number	Viewing angle. The angle from the sensor between nadir (straight down) and the scene center. Measured in degrees (0-90).
eo:azimuth	number	Viewing azimuth angle. The angle measured from the sub-satellite point (point on the ground below the platform) between the scene center and true north. Measured clockwise from north in degrees (0-360).
eo:sun_azimuth	number	Sun azimuth angle. From the scene center point on the ground, this is the angle between truth north and the sun. Measured clockwise in degrees (0-360).
eo:sun_elevation	number	Sun elevation angle. The angle from the tangent of the scene center point to the sun. Measured from the horizon in degrees (0-90).

EO Band Object

Field Name	Type	Description
name	string	The name of the band (e.g., "B01", "B02", "B1", "B5", "QA").
common_name	string	The name commonly used to refer to the band to make it easier to search for bands across instruments. See below for a list of accepted common names.
description	string	Description to fully explain the band. CommonMark 0.28 syntax MAY be used for rich text representation.
gsd	number	Ground Sample Distance, the nominal distance between pixel centers available, in meters. Defaults to `eo:gsd` if not provided.
accuracy	number	The expected error between the measured location and the true location of a pixel, in meters on the ground.
center_wavelength	number	The center wavelength of the band, in micrometers (μm).
full_width_half_max	number	Full width at half maximum (FWHM). The width of the band, as measured at half the maximum transmission, in micrometers (μm).

EO Band Common Names

Common Name	Band Range (µm)	Landsat 5/7	Landsat 8	Sentinel 2	MODIS
coastal	0.40 - 0.45		1	1	
blue	0.45 - 0.50	1	2	2	3
green	0.50 - 0.60	2	3	3	4
red	0.60 - 0.70	3	4	4	1
yellow	0.58 - 0.62				
pan	0.50 - 0.70	8 (*L7 only*)	8		
rededge	0.70 - 0.75				
nir	0.75 - 1.00	4		8	2
nir08	0.75 - 0.90		5	8a	
nir09	0.85 - 1.05			9	
cirrus	1.35 - 1.40		9	10	26
swir16	1.55 - 1.75	5	6	11	6
swir22	2.10 - 2.30	7	7	12	7
lwir	10.5 - 12.5	6			
lwir11	10.5 - 11.5		10		31
lwir12	11.5 - 12.5		11		32

Linking Assets to Bands

```
- eo:bands: [
    - {
        name: "B1",
        common_name: "coastal",
        gsd: 30,
        center_wavelength: 0.44,
        full_width_half_max: 0.02
      },
    - {
        name: "B2",
        common_name: "blue",
        gsd: 30,
        center_wavelength: 0.48,
        full_width_half_max: 0.06
      },
    - {
        name: "B3",
        common_name: "green",
        gsd: 30,
        center_wavelength: 0.56,
        full_width_half_max: 0.06
      },
```

```
- assets: {
    - index: {
        type: "text/html",
        title: "HTML index page"
      },
    - thumbnail: {
        title: "Thumbnail image",
        type: "image/jpeg"
      },
    - B1: {
        type: "image/x.geotiff",
      - eo:bands: [
            0
        ],
        title: "Band 1 (coastal)"
      },
    - B2: {
        type: "image/x.geotiff",
      - eo:bands: [
            1
        ],
        title: "Band 2 (blue)"
      },
    - B3: {
        type: "image/x.geotiff",
      - eo:bands: [
            2
        ],
        title: "Band 3 (green)"
      },
```

STAC Ecosystem

https://github.com/radiantearth/stac-spec/blob/master/implementations.md

STAC Browser

- Browser for navigating static STAC catalogs:
  - https://github.com/radiantearth/stac-browser
- Navigate through Catalogs and Collections, down to Items.

Planet Disaster Data (planet-disaster-data)

Version 1.0.0

https://storage.googleapis.com/pdd-stac/disasters/catalog.json

Planet Disaster Data makes imagery available directly to the public, volunteers, humanitarian organizations, and other coordinating bodies in support of the International Charter for Space and Major Disasters. Data is released for individual disaster events, providing a 30 day window pre- and post-disaster. Imagery is provided under Creative Commons licenses, free of charge, with either CC-BY-SA or CC-BY-NC.

Catalogs

Title
Hurricane Harvey



Provider

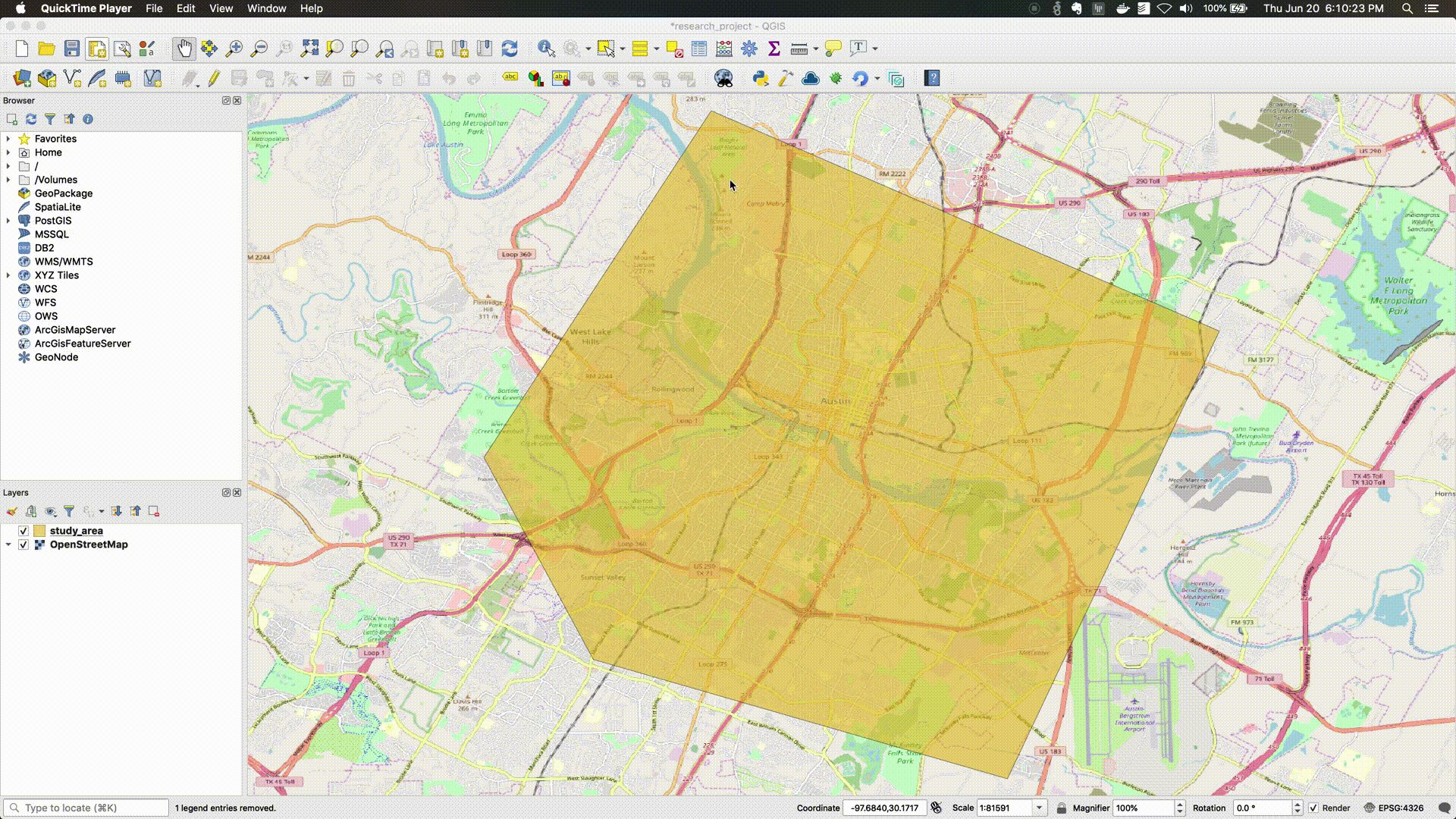
Planet Disaster Team <disaster-team@planet.com> *(producer, licensor)*

STAC Version	0.6.0
Keywords	disaster, open
License	CC-BY-SA-4.0
Temporal Extent	8/28/2017, 11:00:00 AM - now

Powered by STAC Browser

QGIS Plugin

- Search for Items in any publicly available STAC catalog
  - https://github.com/kbgg/qgis-stac-browser
- Show footprints of found Items
- Downloading assets of found Items
- Display assets


QuickTime Player File Edit View Window Help
100% Thu Jun 20 6:10:23 PM
*research_project - QGIS
Browser
Favorites
Home
/
/Volumes
GeoPackage
SpatiaLite
PostGIS
MSSQL
DB2
WMS/WMTS
XYZ Tiles
WCS
WFS
OWS
ArcGisMapServer
ArcGisFeatureServer
GeoNode
Layers
study_area
OpenStreetMap
Type to locate (⌘K)
1 legend entries removed.
Coordinate -97.6840,30.1717
Scale 1:81591
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326

sat-api

https://github.com/sat-utils/sat-api

- STAC dynamic API reference implementation
- Node library for a (STAC) API
- Deploy your own API: https://github.com/sat-utils/sat-api-deployment
- Crawl existing static STAC catalogs to index
- Keep up to date by subscribing to STAC SNS topics

https://earth-search.aws.element84.com/stac

+ - View source


```
{
   stac_version: "0.6.2",
   id: "earth-search",
   title: "Earth on AWS STAC API",
   description: "A STAC API powered by sat-api for Earth on AWS datasets by Element 84",
 - links: [
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-awfi"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan10m"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-mux"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/sentinel-2-l1c"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/landsat-8-l1"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan5m"
       },
     - {
           rel: "self",
           href: "https://earth-search.aws.element84.com/stac"
       }
   ]
}
```


links

sat-stac

https://github.com/sat-utils/sat-stac

- Python library for creating or working with STAC catalogs, collections, and items
- Used for creating existing catalogs, such as
  - https://landsat.stac.cloud
  - https://sentinel.stac.cloud
  - http://cbers.stac.cloud/
- See the Jupyter Notebook tutorials

sat-stac-landsat

https://github.com/sat-utils/sat-stac-landsat

- Indexes Landsat-8 scenes on AWS
- Transform Landsat metadata to STAC (MTL -> STAC)
- Deployed Lambda function for real-time publishing of new Landsat scenes
    - SNS message with entire STAC Item
    - See repo for SNS Topic ARN

sat-stac-sentinel

https://github.com/sat-utils/sat-stac-sentinel

- Sentinel-2(a/b), Sentinel-1 coming
- Index Sentinel scenes on AWS
- Transform Sentinel metadata to STAC (tileInfo.json -> STAC)
- Deployed Lambda function for real-time publishing new Sentinel scenes
    - SNS message with entire STAC Item
    - See repo for SNS Topic ARN

sat-search

https://github.com/sat-utils/sat-search

- Python library for searching STAC complaint endpoints
- Command Line Interface (CLI) as a general purpose tool
- Save search results to be loaded and used later
- Download specific assets from all items
- Specify specific band colors to download those bands
    - e.g., Download all “red” and “nir” bands from these Landsat and Sentinel Items

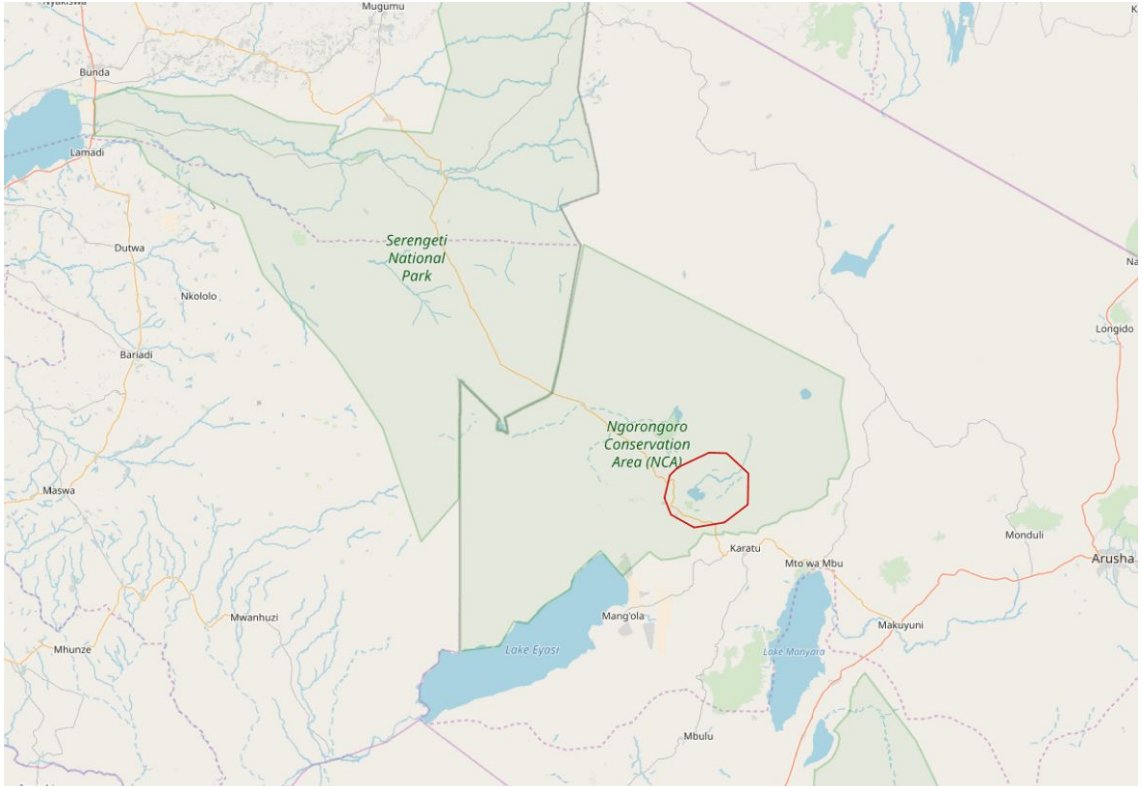
```
$ sat-search search --intersects maine.geojson --found
1674 items found
$ clear
```

sat-fetch

https://github.com/sat-utils/sat-fetch

- Works just like sat-search, plus...
- Option to fetch imagery for just AOIs, rather than the entire asset
- Requires GDAL, or use publicly available Docker image

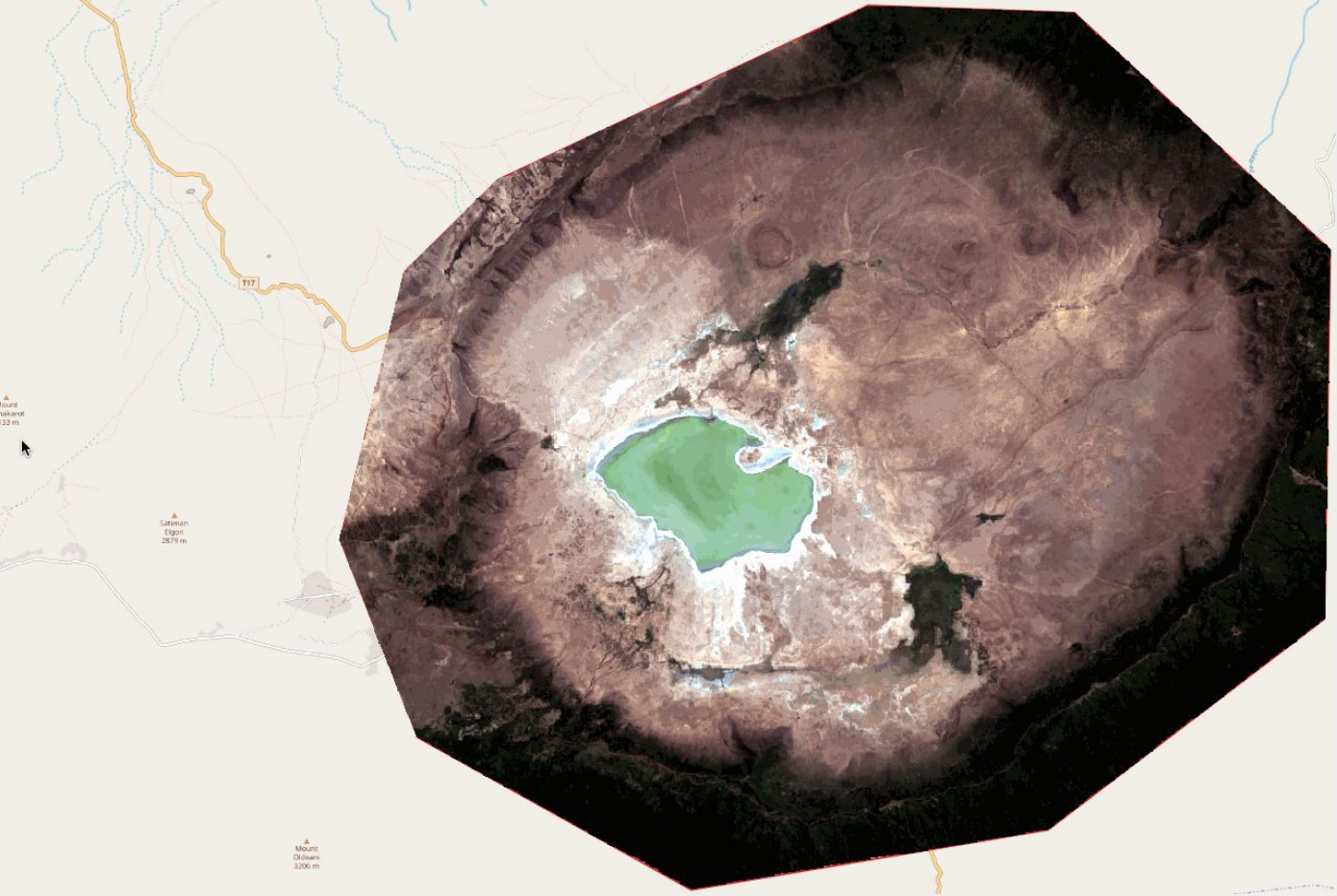
Ngorongoro crater: landsat-8 and sentinel-2



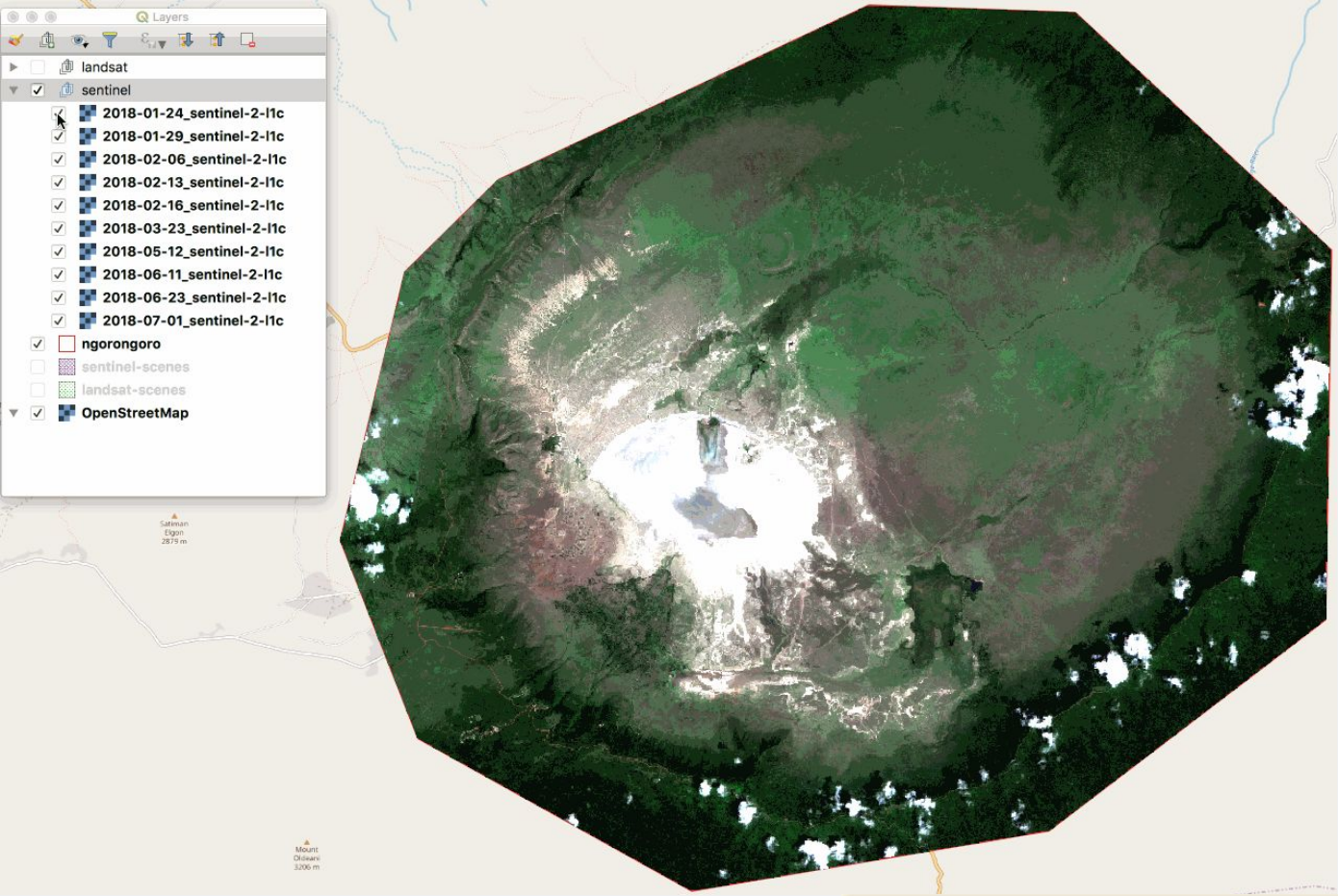
Ngorongoro crater: landsat-8 scenes

```
sat-search search
    \ --intersects ngorongoro.geojson --eo:cloud_cover 0/5 \
    -c landsat-8-l1 --save landsat-scenes.geojson \
    --print_md date eo:cloud_cover id
```

```
2014-12-11  5  LC08_L1TP_169062_20141211_20170416_01
2015-01-28  1  LC08_L1TP_169062_20150128_20170413_01
2015-02-13  0  LC08_L1TP_169062_20150213_20170413_01
2015-08-24  2  LC08_L1TP_169062_20150824_20170405_01
2015-09-09  1  LC08_L1TP_169062_20150909_20170404_01
2015-12-30  0  LC08_L1TP_169062_20151230_20170331_01
2016-02-16  0  LC08_L1TP_169062_20160216_20180526_01
2016-08-10  2  LC08_L1TP_169062_20160810_20170322_01
2016-10-13  0  LC08_L1TP_169062_20161013_20180526_01
2016-10-29  3  LC08_L1TP_169062_20161029_20170319_01
2017-01-17  0  LC08_L1TP_169062_20170117_20170311_01
2017-11-17  2  LC08_L1TP_169062_20171117_20171122_01
2018-02-05  0  LC08_L1TP_169062_20180205_20180221_01
2018-05-28  3  LC08_L1TP_169062_20180528_20180605_01
18 scenes found
```




```
$ time sat-fetch load landsat-scenes.geojson \
    --download red green blue
real 0m59.674s
```




```
$ time sat-fetch load sentinel-scenes.geojson \
    --download red green blue
real 8m6.525s
```

sat-api-browser

https://github.com/sat-utils/sat-api-browser

- Initial prototype
- Web-based UI for searching and saving results from STAC APIs
- Works with any STAC compliant API

SAT-API Browser

BUILD QUERY | RESULTS | CANDIDATES

Date Range

Start Date (required): 2019/07/14 06:33 PM

End Date (required): 2019/07/15 06:33 PM

Area Of Interest (required)

DRAW BBOX ON MAP

Property Filters

Add new filter

Select a property to filter

SUBMIT QUERY



intake-stac

https://github.com/pangeo-data/intake-stac

- Plugin to Intake
- Turn STAC Item Assets into Dask arrays
- Stack together multiple Assets
- Load STAC search results as Intake catalogs (in 0.2.0)

NASA CMR STAC

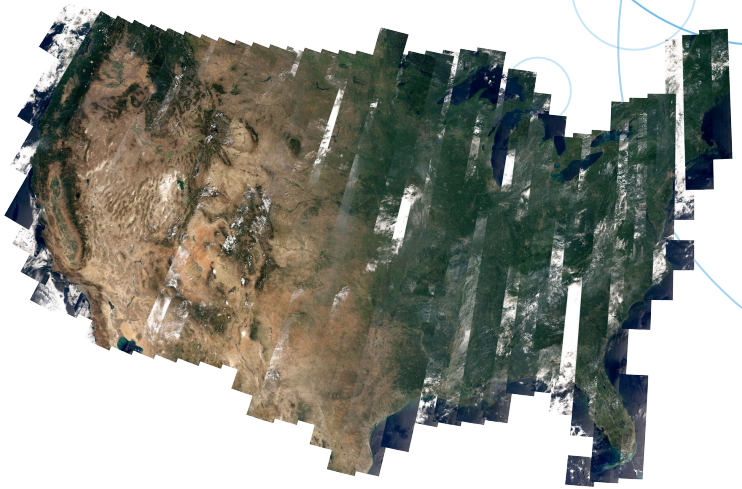
- NASA Common Metadata Repository
  - All NASA data from all DAACs in CMR … (?)
- Element 84 currently working on adding STAC support to CMR
  - By October
- Implemented as a proxy to CMR
  - Separate STAC API endpoint
  - no ingestion
  - Targeting STAC 0.8.0 (to be released by end of month)

E84 Earth Search

https://earth-search.aws.element84.com/stac

Public Search and Discovery API for Earth on AWS Datasets

- Central search catalog and standardized metadata
- Single API that allows searching of STAC metadata for all Earth on AWS datasets
- Powered by sat-api (https://github.com/sat-utils/sat-api)
- Backed by an ElasticSearch index
- Proposed initial datasets
  - Sentinel-1
  - Sentinel-2
  - CBERS-4
  - Landsat-8
  - NAIP



Landsat-8

Landsat is perhaps one of the most oft-cited and used EO datasets because of its historical archive (Landsat-8 was launched in 2013, but the entire Landsat archive goes back to 1982, with some caveats).

For updates, releases, news:

@GeoSkeptic

Element 84

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